SIKSHA 'O' ANUSANDHAN (DEEMED TO BE UNIVERSITY)
Institute of Technical Education and Research | Faculty of Engineering & Technology
Department of Computer Science & Engineering

CSE 4571 – PRACTICAL ROBOTICS PROJECTS WITH ARDUINO

EXPERIMENT 2
ANALOG SENSOR INTERFACING AND REAL-TIME DATA ACQUISITION

Duration	Level	Mode	CO Mapping
2 Hours	B.Tech CSE Final Year	Hardware / Wokwi	CO1, CO2, CO5

1. Aim of the Experiment

To interface an analog sensor with Arduino UNO, acquire ADC data and study its response to changing illumination.

2. Learning Objectives

- Understand ADC-based acquisition.
- Read a Wokwi photoresistor.
- Observe sensor response at different lux values.
- Use Serial Monitor/Serial Plotter for visualization.

3. Components / TinkerCAD or Wokwi Parts

Component	Quantity	Purpose
Arduino UNO	1	Microcontroller / ADC
Photoresistor module	1	Analog illumination sensor
LED	1	Threshold indication
220 Ω resistor	1	LED current limiting
Breadboard	1	Circuit assembly
Jumper wires	As required	Interconnections

4. Theory

Arduino UNO provides 10-bit analog inputs. The Wokwi photoresistor module exposes AO, whose voltage varies with simulated illumination. This experiment introduces acquisition, thresholding and simple calibration.

ADC value = analogRead(A0) → 0 to 1023

Approx. voltage = ADC × 5.0 / 1023.0

5. Pre-Lab Questions

1. What is ADC resolution?

2. What does analogRead() return?

3. Why is calibration important?

4. Why is A0 used for the photoresistor signal?

5. Circuit Diagram

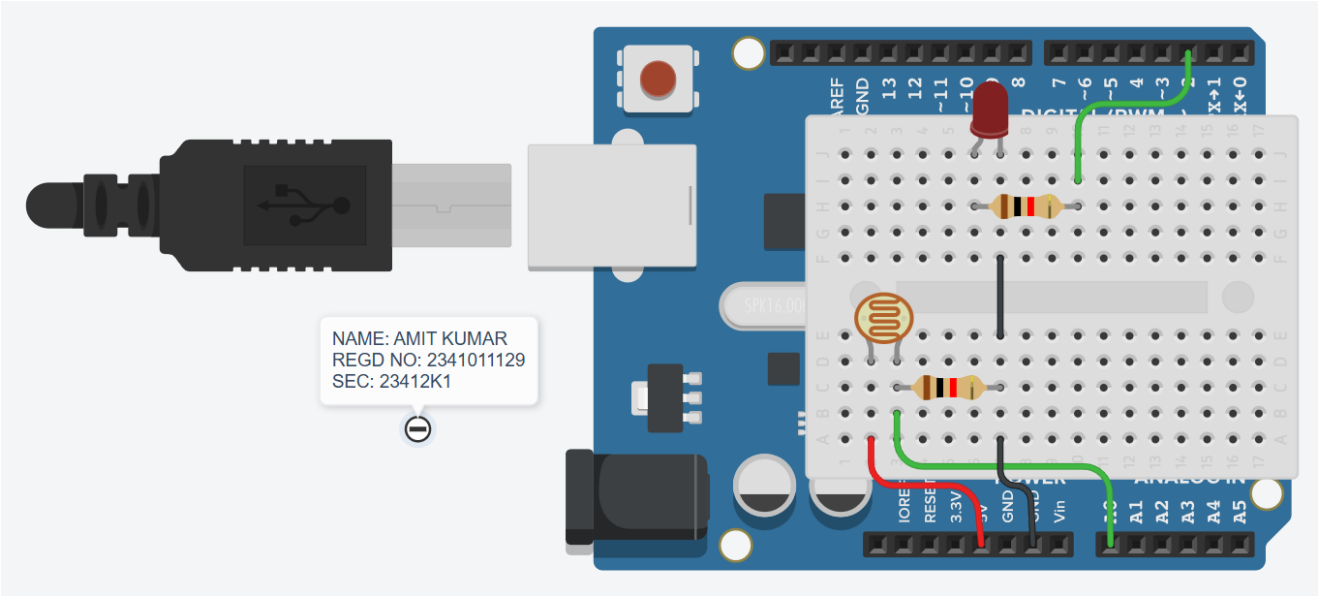


Figure 1. Arduino UNO, TinkerCAD/Wokwi photoresistor module and LED threshold indicator.

6. Connection Table

Component / Signal	Arduino Pin	Connection Details
Photoresistor AO	A0	Analog input
Photoresistor VCC	5V	Supply
Photoresistor GND	GND	Common ground
LED Anode (+)	D2	Through 220 Ω resistor
LED Cathode (−)	GND	Ground

7. Circuit Diagram Implemented Using Hardware's

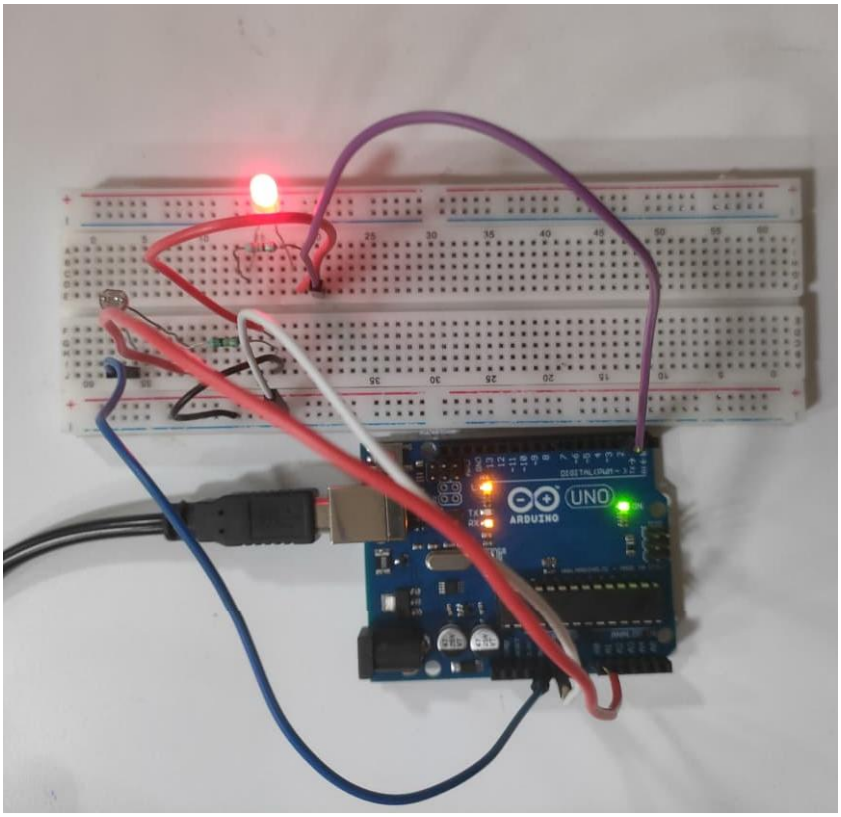


Figure 2. Hardware Implementation of Arduino UNO, TinkerCAD/Wokwi photoresistor module and LED threshold indicator.

8. Procedure

- Add the photoresistor and LED to the Arduino UNO Wokwi project.
- Connect AO pin of photoresistor module to A0 of Arduino Uno and the LED to D2 through a 220 Ω resistor.
- Complete 5V and GND connections and verify common ground.
- Read A0 and calculate the approximate voltage.
- Change the simulated illumination/lux value while the simulation is running.
- Record ADC values and observe the trend in Serial Monitor/Serial Plotter.
- Take at least five readings and record the LED state.
- Plot or inspect ADC against simulated illumination.

9. Arduino Program

```
const int S = A0;
const int LED = 2;
void setup()
{
  Serial.begin(9600);
  pinMode(LED, OUTPUT);
}
void loop()
{
  int a = analogRead(S);
  float v = a * 5.0 / 1023.0;
  Serial.print("ADC=");
  Serial.print(a);
  Serial.print(" voltage=");
  Serial.println(v, 2);
  digitalWrite(LED, a < 500);
}
```

10. Observation Table

Sl. No.	Simulated Lux	ADC Value	Voltage (V)	LED State
1	N/A (Tinkercad)	70	0.34	ON
2	N/A (Tinkercad)	278	1.36	ON
3	N/A (Tinkercad)	439	2.15	ON
4	N/A (Tinkercad)	552	2.70	OFF
5	N/A (Tinkercad)	679	3.32	OFF

11. Expected Output / Analysis

The Serial Monitor displays the ADC value and the corresponding calculated voltage. The LED is switched ON when the ADC reading is below 500 and OFF otherwise.

ADC Condition	LED	Interpretation
ADC < 500	ON	Below programmed threshold
ADC ≥ 500	OFF	At/above programmed threshold

12. Result

The analog photoresistor was successfully interfaced with Arduino UNO. ADC data and corresponding sensor voltage were acquired, and the LED threshold response was verified in Wokwi.

13. Post-Lab / Application Questions

5. Plot ADC against simulated illumination and describe the trend.

6. Change the threshold and compare the results.

7. Why can sensor response be nonlinear?

8. How could this sensor be used for automatic lighting in a robot or smart environment?

14. Precautions

- Verify pin assignments before running the simulation.
- Use common ground for connected devices.
- Record both the input condition and observed output.
- For physical implementation, consider current, voltage, mechanical loading and sensor tolerances.

15. Suggested Extension

Modify the program to add one meaningful feature, document the change, and preserve the original behaviour as a baseline.

```
const int S = A0;
const int LED = 2;
void setup() {
  Serial.begin(9600);
  pinMode(LED, OUTPUT);
}
void loop() {
  int a = analogRead(S);
  float v = a * 5.0 / 1023.0;
  Serial.print("ADC=");
  Serial.print(a);
  Serial.print(" voltage=");
  Serial.println(v, 2);
  digitalWrite(LED, a < 500);
  if (a < 500) {
    Serial.println("Status: LOW LIGHT");
  }
```

```
  else {
    Serial.println("Status: SUFFICIENT LIGHT");
  }
  delay(500);
}
```

Suggested Extension:

Added a light-status message to the Serial Monitor. The program displays "LOW LIGHT" when ADC < 500 and "SUFFICIENT LIGHT" when ADC ≥ 500. The original LED threshold operation is preserved.

Result of Extension:

The additional light-status information was successfully displayed on the Serial Monitor without changing the original LED behavior.

16. Student Record

Student Name	Registration No.	Date	Faculty Signature
Amit Kumar	**2341011129**		